

NUX IVE - a research tool for comparing voice user interface and graphical user interface in VR

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ABSTRACT

A trend of using natural interaction such as speech is clearly visible in human-computer interaction, while in interactive virtual environments (IVE) still it has not become a common practice. Most of input interface elements are graphical and usually they are implemented as non-diegetic 2D boards hanging in 3D space. Such holographic interfaces are usually hard to learn and operate, especially for inexperienced users. We have observed a need to explore the potential of using multimodal interfaces in VR and conduct the systematic research that compare the interaction mode in order to optimize the interface and increase the quality of user experience (UX). We introduce a new IVE designed to compare the user interaction between the mode with traditional graphical user interface (GUI) with the mode in which every element of interface is replaced by voice user interface (VUI). In each version, four scenarios of interaction with a virtual assistant in a sci-fi location are implemented using Unreal Engine, each of them lasting several minutes. The IVE is supplemented with tools for automatic generating reports on user behavior (clicktracking, audiotracking and eyetracking) which makes it useful for UX and usability studies.

Index Terms: Virtual reality—Voice user interface—Holographic user interface—Human-computer interaction; Virtual assistant—Multimodal interface—Conversational agent

1 INTRODUCTION

In human-computer interaction (HCI) using natural interfaces as the form of communication which are more human-like (e.g. speech, gestures or emotions) have a strong potential to help developing more effective, intuitive and user-friendly systems [6]. Traditional graphical interfaces are often a barrier in using head mounted display (HMD) systems and instead of supporting user experience - they induce a steep learning curve, especially for less experienced users. Instead, voice interaction could be more immersive and faster in both onboarding and regular usage of VR applications. What is more, it allow focusing more on the game world instead of non-diegetic interface controls. With the growing precision of automatic speech recognition (ASR) systems, voice interaction is widely adapted by



Figure 1: GUI version of the hints given by the assistant in the Scenario 2 - searching for boxes with fruits

users in recent years, starting from interactive voice response (IVR) systems, personal voice assistants on Android smartphones (Google Assistant, Bixby) or iPhones (Siri) or operation systems (Cortana), ending with smart speakers. While VUI have several undoubted advantages over GUI, like independence from hands and particular space points / touch points, there are some challenges in adopting this innovation by more sceptical group of users (like earlier habits or reluctance to speak to AI). Also while comparing these two types of interaction, new challenges arise. Recently, some new approaches to comparison schemes were introduced ([2]). In the NUX project we follow recommendations to triangulate subjective declarative measures like questionnaires [1] or qualitative methods ([5] like in-depth interviews with objective methods like eye-tracking [3] as a mixed approach (e.g. [4]).

The NUX project aims at exploring the potential of voice interaction in the VR, for both games and training simulators. In order to investigate user interaction and preferences, we have constructed the NUX IVE - a research scenarios and a virtual environment supplemented with the tools for automatic reporting the data on user behavior.

2 SOLUTION

2.1 Environment and interaction

The environments is designed as a futuristic office at the space station. At the office there are several locations, e.g. work zone, battery charging zone, chillout zone or a canteen. Other workers are robots or people (non-player characters). The user is having their first day at work in the office and they are supposed to get acquainted with the new role, duties and the company with help of the virtual

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Figure 2: GUI version of the input of the user name in the Scenario 3 - arrangement of the desk



Figure 3: VUI version of the virtual assistant in a Scenario 3 - arrangement of the desk

assistant. The assistant is a robot named Bobby (a name chosen as neutral and easy to spell) and it is designed to have genderless look combined with blue and pink elements. The convention was chosen after the initial quantitative pre-test ($N=63$) in which respondents decided in which location and style they would prefer to learn soft skills (there were several options, e.g. historical or contemporary locations). In the GUI version, both input and output of the system was implemented graphically in the form of 2D buttons and labels (see Fig. 1 or a virtual keyboard (Fig. 2). In the VUI version, Bobby speaks to the user and user voice commands are the input to the system, there are no GUI elements (Fig. 3). User is allowed to use spontaneous natural language so the conversation can be more human-like.

2.2 Scenarios

Each variant of the IVE (VUI and GUI version) contains 4 parallel scenarios, which differ only in type of interaction with the virtual assistant. The scenarios were designed to test various functionalities of the assistant (a source of knowledge, a guide, an inventory). In Scenarios 1 and 4 the user decides on the duration of the scenario, while in Scenario 2 it is determined by finishing the task and in Scenario 4 it is fixed by number of locations. **Scenario 1** is a free chatting with the assistant on the topics e.g. connected to the new job, office, or the assistant itself. **Scenario 2**: is a searching for 4 fruit boxes scattered around the office; Bobby gives hints where to find them (see Fig. 1). **Scenario 3** is an arranging the desk for work; assistant plays the role of inventory and gives items to the user (see Fig. 3). **Scenario 4** is based on visiting various locations in the office. The assistant shows user around the office and asks questions checking if everything was understood.

2.3 Technical description

The IVE was implemented in Unreal Engine. It can be used with HTC Vive Pro Eye headset and Vive 2.0 controllers, configured for the roomscale experience. The built-in microphone is used as an audio input. Before the usage of the IVE, a chaperrone zone should

be set up and the eyetracker embedded in the headset should be calibrated.

2.4 Research tools

Additionally, we have implemented functionalities that allow acquisition and storing user behavior data in csv format. **Clicktracking** report is documenting usage of each of the controller buttons in particular timestamps. **Eyetracking** report supports investigating the patterns of user attention drawn by two areas - the assistant head and body; start of looking and looking duration is reported and the total time of looking is counted. **Audiotracking** report collects transcripts of each of the utterance of the user as well as assistant response and the timestamps.

The environment was developed in an iterative lean process using user-centered design paradigm. In the initial phase, a Wizard of Oz method was used to design the dialog system. After each sprint, usability test were conducted with at least 5 users according to the Nielsen model. The final version was tested with 16 users using qualitative approach (2-3 h per user).

3 DISCUSSION AND FUTURE WORKS

The NUX IVE tool can be applied for exploring the impact of the conversational interface on the ergonomics and usability as well as user experience of the virtual environments. Adding the adequate questionnaire (e.g. immersion, presence or system usability) or user characteristics (e.g. personality, mood or digital competences levels) to the research study can bring additional information on the factors that impacts preferences of each users towards the mode of interaction. What is more, the construction of the scenarios (parallelism of all the functionalities except the interface mode) can bring valuable insights concerning the strengths and limitations of voice and graphical interaction and choose the optimal fusion of both modalities. The IVE architecture is modular and easy to be supplemented with further scenarios. We are currently adding functionalities allowing for examining various forms and features of the virtual assistant (e.g. humanoid, animal, cartoon or a robot) with different voice timbre or compare communication in additional languages. The test environment is planned to be released at the end of the project together with the creator allowing to design and adjust the VUI.

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